Utilizing Accelerometer and Photoplethysmography Data from A Smartwatch for the Quantitative Detection of Adult Sleep Apnea

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Abstract

Sleep apnea (SA) is linked to cardiovascular disease and is marked by insufficient global awareness and low diagnostic rates. The implementation of effective screening tools has the potential to enhance early diagnosis, raise public awareness, and improve the overall health and quality of life for patients. This study aimed to evaluate the capability of the S7 smartwatch, utilizing accelerometry and photoplethysmography, to quantify apnea-hypopnea events during overnight monitoring. Clinical findings demonstrate a strong correlation between the apnea-hypopnea index (AHI) estimated by the S7 smartwatch and by home sleep testing (R=0.88). The accuracy rates for detecting moderate SA (AHI $\geq$ 15) and severe SA (AHI $\geq$ 30) using smartwatch-based measurements are 81% and 97%, respectively. Consequently, our findings indicate that the smartwatch can effectively serve as reliable tools for adult SA screening.

CCS Concepts

• **Applied computing**; • **Life and medical sciences**; • **Consumer health**;

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ICCBB 2024, Kyoto, Japan

ACM ISBN 979-8-4007-0962-3/2024/11
https://doi.org/10.1145/3715020.3715066

Keywords

Sleep apnea, AHI, smartwatch, accelerometer,photoplethysmography

ACM Reference Format:
Yabin Meng, Wentian Jiang, Ruiqi Guo, Peng Deng, Zhao He, Shunde Hong, and Chi-Wei Chang. 2024. Utilizing Accelerometer and Photoplethysmography Data from A Smartwatch for the Quantitative Detection of Adult Sleep Apnea. In *2024 8th International Conference on Computational Biology and Bioinformatics (ICCBB 2024), November 28–30, 2024, Kyoto, Japan.* ACM, New York, NY, USA, 6 pages. https://doi.org/10.1145/3715020.3715066

1 Introduction

Sleep apnea (SA) is a prevalent sleep disorder linked to numerous adverse health outcomes, such as cardiovascular disease [1, 2], cognitive decline [3] , and an elevated risk of accidents attributable to excessive daytime sleepiness [4]. Current estimates suggest that over 80% of individuals with sleep apnea globally remain undiagnosed [5, 6], underscoring the critical need to enhance awareness and improve diagnostic rates.SA is associated with cardiovascular disease due to factors like nighttime oxygen deprivation, increased carbon dioxide levels, and blood pressure changes. Research shows that SA patients are at greater risk for hypertension, coronary artery disease, arrhythmia, and heart failure. [6] SA is also associated with cognitive issues like poor attention, memory, and executive function, likely from disrupted sleep and low nighttime oxygen. Daytime sleepiness further impacts quality of life and increases accident risks.

Therefore, the development of a widely accepted and effective screening tool is essential for enhancing the diagnostic rate of sleep apnea. Such a tool would facilitate early detection, enabling timely intervention and treatment, thereby mitigating the risk of complications. Early intervention has the potential to alleviate the economic burden associated with prolonged medical expenses and the deterioration in quality of life. Furthermore, this approach contributes to increasing public awareness of sleep apnea and promotes the adoption of beneficial lifestyle modifications, such as weight loss and smoking cessation. These changes can mitigate the risk of associated illnesses and enhance the overall health and quality of life for patients.

To enhance sleep apnea diagnosis, researchers are leveraging wearable technology. Smartwatches equipped with accelerometers (ACC) and photoplethysmography (PPG) offer a low-power, long-term monitoring solution. These devices can screen for sleep apnea by analyzing pulse and heart rate variations, along with frequency and amplitude modulation characteristics. However, the clinical validation of smartwatch-based ACC and PPG signals for sleep respiratory screening remains limited, and the specific algorithms and methodologies employed are not comprehensively disclosed in the existing literature. [7, 8]

This article introduces a methodology for extracting heart rate and respiratory signals from smartwatches to identify sleep apnea events and calculate the Apnea-Hypopnea Index (AHI). It assesses the concordance of AHI measurements obtained from the smartwatch with those derived from a home sleep test (HST) device. By comparing the reference AHI from the HST with the AHI derived from the smartwatch, Receiver Operating Characteristic(ROC) curve is employed to validate the efficacy of the S7 smartwatch (SKG Health Technologies, China) as a screening tool for sleep apnea.

2 Methods

2.1 Subjects and Protocol

The mean age of the 35 participants was 48 years, with an age range spanning from 28 to 68 years. These participants underwent nocturnal HST between April 2024 and May 2024. The participants were enrolled from the Chinese Academy of Medical Sciences Fuwai Shenzhen Hospital (IRB approval number: SP2023144-01-A). During the testing phase, each participant was mandated to utilize the HST device SOMNO screen™ Plus (Somno GmbH, Germany) in conjunction with wearing an S7 smartwatch. The comprehensive demographic characteristics of the patients, as well as the specifics of the HST protocol, have been documented in prior research studies. [9]

Previous research has concentrated on the continuous monitoring of respiratory rates and the subsequent analysis of its accuracy. In the present study, we developed a method to identify apnea-hypopnea events utilizing ACC and PPG data from the S7 smartwatch, and employed this method to assess the AHI values. Following an 8-hour period of continuous nocturnal monitoring, the standard AHI value was derived from a HST device, which evaluates physiological signals and identifies apnea-hypopnea events in accordance with the guidelines established by the American Academy of Sleep Medicine (AASM). [10] The severity of sleep apnea (SA) is classified into four distinct categories according to the Apnea-Hypopnea Index (AHI): normal (AHI $<$ 5), mild (5 $\leq$ AHI $<$ 15), moderate (15 $\leq$ AHI $<$ 30), and severe (AHI $\geq$ 30).

2.2 Data processing and feature extraction

Throughout the night, ACC and PPG signals were recorded at a sampling frequency of 100 Hz. Given that the S7 smartwatch lacked a SA screening function for the duration of this study, the raw 3-axis ACC and PPG data were transmitted to a computer via Bluetooth, using the software development kit (SDK) version W25.0. To detect sleep apnea and hypopnea events using s7 smartwatch sensors, we selected PPG features and ACC features correlated respectively with heart rate variability (HRV) and respiratory effort. The reduction in blood oxygen levels resulting from respiratory events can induce alterations in HRV, manifesting as increased heart rate variability and modifications in the distribution of energy across frequency bands. Furthermore, minor body movements within a specific frequency band, induced by respiratory effort, can be detected through the accelerometer signal embedded in the smartwatch. Consequently, the respiratory rate and amplitude of respiratory effort can be quantified based on this data. However, since accelerometers also capture other types of body movements, sophisticated algorithms are necessary to differentiate respiratory motion from other activities. In this study, we introduced three smartwatch-based methods to identify apnea-hypopnea events: the photoplethysmography (PPG) method, the accelerometer (ACC) method, and the fusion method. The details of these methods are described as follows:.

2.3 PPG method

Heart rate and HRV can be quantitatively assessed through the analysis of PPG signals utilizing peak detection algorithms. Apnea and hypopnea events, which are frequently associated with an elevation in heart rate.For evaluation purposes, we selected two HRV features that exhibit a high correlation with respiratory events: the very low frequency ratio (VLF ratio) and the standard deviation of the NN intervals (SDNN). The VLF ratio quantifies the proportional relationship between the energy in the 0-0.04 Hz frequency band and the energy in the 0-0.5 Hz frequency band within a one-minute PPG signal.On the other hand, SDNN quantifies the standard deviation of intervals between successive normal heartbeats (NN intervals).

We compute the SDNNfrom PPG signals within a one-minute duration. To adjusting individual variability in SDNN values, we determine the median value of the normal heart beat interval for each individual over the course of the night, which serves as the normalization reference (NN_ref). The normalized SDNN for an individual (SDNN_nor) is then calculated using the following formula:

$$SDNN\_nor = SDNN/NN\_ref \tag{1}$$

In the PPG method, we utilized two features, SDNN_nor and the VLF ratio, along with their respective thresholds to identify apnea-hypopnea events and to calculate the smartwatch-derived AHI.

2.4 ACC method

Nighttime sleep movements can be tracked by measuring the ACC signal's root mean square sum of three-axis signals surpassing a threshold 40 mg. Besides the episode annotated the sleep movements, we set a 6-second time window and a 3-second sliding step to calculate the variance of the x-axis, y-axis, and z-axis to evaluate the intensity of respiratory effort within the window.The amplitude and frequency modulation signals of ACC related to respiratory effort can be extractedfrom frequency range of 0.15-0.45 Hz.

Nocturnal sleep movements can be monitored by assessing the root mean square (RMS) sum of three-axis accelerometer (ACC) signals that exceed a threshold of 40 mg. In addition to annotating episodes of sleep movements, we employed a 6-second time window with a 3-second sliding step to compute the variance of the x-axis, y-axis, and z-axis signals, thereby evaluating the intensity of respiratory effort within each window. The amplitude and frequency modulation signals of the ACC, which are associated with respiratory effort, can be extracted from the frequency range of 0.15 to 0.45 Hz. For each successive time interval, the respiratory amplitude (RM) is quantified using ACC signals as follow:

$$RM = \sum_{f=0.15}^{0.45} \left(X_f^2 + Y_f^2 + Z_f^2 \right) \tag{2}$$

where that X_f, Y_f, Z_f denote the amplitude at frequency f.

Segments demonstrating substantial increases in RM were identified as potential apnea and hypopnea events. The AHI value was subsequently calculated, representing the number of respiratory events per hour of sleep.

2.5 Fusion method

The fusion algorithm functions by concurrently detecting elevations in HRV and RM, employing relatively lenient thresholds for both parameters. Apnea and hypopnea events are identified when both HRV and RM meet the predefined criteria for these repspiratory events within a 1-minute window.

2.6 Statistical Validation

To assess the diagnostic capability of smartwatches in identifying SA of varying severity, the AHI derived from the smartwatch (AHI_{watch}) was compared to the reference AHI obtained from theHSTdevice (AHI_{ref}). A receiver operating characteristic (ROC) curve was generated based on this comparison. The area under the curve (AUC) of the ROC plot, along with its 95% confidence interval, was computed to assess three distinct AHI_{watch} values derived from various smartwatch-based algorithms utilizing PPG, ACC, and fusion methods. In addition, the agreement between AHI_{watch} and AHI_{ref} was assessed using the Pearson correlation coefficient and Bland-Altman plot. We also evaluated the performance of the S7 smartwatch in screening for moderate and severe SA using sensitivity and specificity metrics. The chi-square test and Cohen's Kappa Statistic were utilized to evaluate the correlation and agreement, respectively. Statistical significance was established by employing a threshold criterion of $P < 0.05$.

3 Results

3.1 Capture of respiratory signals using the HST device and the S7 smartwatch.

Figure 1 displays representative 2-minute segments of both apnea events and normal episodes, featuring synchronized signals obtained from the HST device and the S7 smartwatch. Respiratory signals and events were monitored during sleep using a HST device, which recorded nasal airflow, abdominal and thoracic movements, and blood oxygen levels. Concurrently, the S7 smartwatch internally acquired PPG and three-axis ACC signals, synchronizing these data streams with the HST device.

3.2 Subjects' characteristics

HST devices were used to assess SA severity in 35 participants: 12 were normal, 12 had mild SA, 4 had moderate SA, and 7 had severe SA. (Normal : $AHI < 5$; mild: $5 \leq AHI < 15$; moderate: $15 \leq AHI < 30$; severe: $AHI \geq 30$) The baseline characteristics of patients, categorized according to the severity level of SA, are detailed in Table 1. There are no statistically significant differences in age and body height among the groups. Statistically significant difference in BMI were observed between the normal-to-mild SA groups and the moderate-to-severe SA groups. ($P<0.01$)

During the synchronization of sensor signals among a total of 35 participants, 3 participants were unable to match the HST device's time flag with the S7 smartwatch, and 1 participant's PPG data was missing. Consequently, only 31 participants had complete synchronized data from both the HST device and the S7 smartwatch. The subsequent analysis focused on the records of these 31 participants.

3.3 AHI_{watch} Validation by HST device

In this article, AHI_{watch} refers to the AHI value calculated using a smartwatch-based algorithm, whereas AHI_{ref} denotes the directly measured AHI value obtained from the standard HST device, serving as the reference value. Figure 2 compares the performance of smartwatch-based SA diagnosis at different AHI_{ref} thresholds ($AHI_{ref} \geq 5$, $AHI_{ref} \geq 15$, $AHI_{ref} \geq 30$). Three ROC from sub Figure 2(a) to Figure 2(c), each corresponding to a different AHI_{ref} threshold, are employed to represent SA diagnosis as determined by HST in comparison to the results obtained from smartwatch sensors. In each ROC sub figure, the AHI_{watch} values generated by ACC or PPG method, as well as those generated by a fusion method (ACC+PPG), are analyzed separately. The AUC and the 95% confidence intervals are presented for each method. In Figure 2(c), the results demonstrated that the diagnostic performance for severe SA using a smartwatch-based algorithm was excellent, as evidenced by AUC values of 0.98 and 0.97 for the ACC and fusion methods, respectively. In contrast, the diagnostic performance for mild SA was suboptimal, as indicated by AUC values below 0.8 across all types of smartwatch-based methods, as illustrated in Figure 2(a).

To conduct a more in-depth analysis, we employed a fusion method to further examine the correlation between AHI values derived from smartwatch-based and HST-based methods. The results revealed a strong correlation ($R=0.88$) between AHI_{ref} and AHI_{watch} (Figure 3). The Bland-Altman plot indicated good agreement for lower AHI_{watch} values; however, it demonstrated a significant underestimation when AHIwatch values exceeded 30, compared to the AHIref values. Despite this underestimation, the performance for diagnosing severe SA remained unaffected. (Figure 3)

In the assessment of patients with moderate-to-severe SA ($AHI \geq 15$), the sensitivity and specificity of the smartwatch-based assessment utilizing the fusion method were 73% and 85%, respectively. (Table 2) The chi-square statistic of 10.3 suggests a significant correlation between the classifications produced by the two devices (P

Table 1: Baseline Characteristics of Patients Stratified by Severity Level of SA (n=35).

Characteristic	Normal	Mild	Moderate	Severe
Age (years old); mean (SD)	47.7 (9.2)	49.7 (10.8)	44.6 (6.7)	46.4 (12.1)
Male, n (%)	9 (81.8)	10 (83.3)	4 (80.0)	7 (100.0)
Height (m); mean (SD)	169 (7)	170 (7)	168 (11)	169 (6)
Weight (kg); mean (SD)	72 (10)	75 (16)	85 (9)	84 (14)
BMI (kg/m2)	25.4 (3.4)	25.7 (4.3)	30.0 (1.0)	29.3 (3.1)
AHI; mean (SD)	2.6 (1.7)	9.4 (2.7)	20.0 (4.2)	50.9 (13.3)
SA: Sleep Apnea; Normal (AHI < 5);Mild (5 ≤ AHI < 15); Moderate (15 ≤ AHI < 30);Severe (AHI ≥ 30)				

Table 2: The confusion matrix for the ACC+PPG model, with classification corresponding to an AHI threshold of 15.

	$AHI_{watch} \geq 15$	$AHI_{watch} < 15$	Total
$AHI_{ref} \geq 15$	8	3	11
$AHI_{ref} < 15$	3	17	20
Total	11	20	31

Table 3: The confusion matrix for the ACC+PPG model, with classification corresponding to an AHI threshold of 30.

	$AHI_{ref} \geq 30$	$AHI_{ref} < 30$	Total
$AHI_{watch} \geq 30$	6	0	6
$AHI_{watch} < 30$	1	24	25
Total	7	24	31

< 0.01). Furthermore, the Cohen's kappa coefficient for the two devices in diagnosing moderate-to-severe sleep apnea was calculated to be 0.577 (95% CI: 0.276-0.878), indicating moderate agreement between the S7 smartwatch and the HST device in the screening of moderate-to-severe SA.

In the evaluation of patients diagnosed with severe sleep apnea (AHI ≥ 30), the sensitivity and specificity of the smartwatch-based assessment utilizing the fusion method were 86% and 100%, respectively. (Table 3)The chi-square statistic of 25.51 indicates a significant correlation between the classifications of the two devices (P < 0.001). The Cohen's kappa coefficient was 0.903 (95% CI 0.716-1.000) for the two devices in diagnosing severe SA, demonstrating substantial agreement between the S7 smartwatch and the HST device in screening for severe SA.

4 Discussion

In this study, we introduced a multi-sensor-based methodology for the convenient estimation of SA utilizing the S7 smartwatch and subsequently validated its concordance with a standard HST device. Several types of physiological parameters and features were analyzed, including HRV, respiratory effort, deep breathing motion, and sleep movement. In comparison to the smartwatch-based model that exclusively employed the ACC sensor and a patent method, which achieved an AUC of ROC of 0.919 and an accuracy rate of 85% for AHI ≥ 30 [11], our results exhibited superior performance and improved AHI resolution, with an AUC of ROC of 0.97 and an accuracy rate of 97% for AHI ≥ 30.

In the fusion method, an increase in HRV serves as a compensatory mechanism in response to oxygen deprivation, which can aid in the identification of respiratory events. Furthermore, specific sleep postures may result in the degradation or complete loss of respiratory signals in wrist ACC data [7]. Consequently, a fusion-based approach can mitigate these deficiencies, thereby maintaining performance and balancing sensitivity and specificity in screening for moderate to severe conditions (Figure 2.b). To establish the robustness of the fusion method's performance, larger sample sizes or multi-night testing are necessary.

Smartwatches equipped with PPG and ACC sensors are capable of monitoring health status during sleep with minimal burden and disturbance. In this study, the proposed approach has shown significant potential as an early screening tool for the diagnosis of SA. However, the concordance of the AHI between data obtained from smartwatches and HST devices necessitates further enhancement. Thus, for patients who screen positive for suspected sleep apnea using smartwatches, confirmation through either HST device or polysomnography is necessary. Furthermore, the limited diagnostic performance for patients with mild SA constrains the effectiveness of smartwatch-based SA screening. This necessitates the development of more sophisticated algorithms to enhance the resolution within the 0-15 AHI range.

5 Conclusion

This study employs ACC and PPG signals from the S7 smartwatch to estimate the severity of SA. The proposed methodologies exhibit promising results in identifying patients with severe SA, achieving

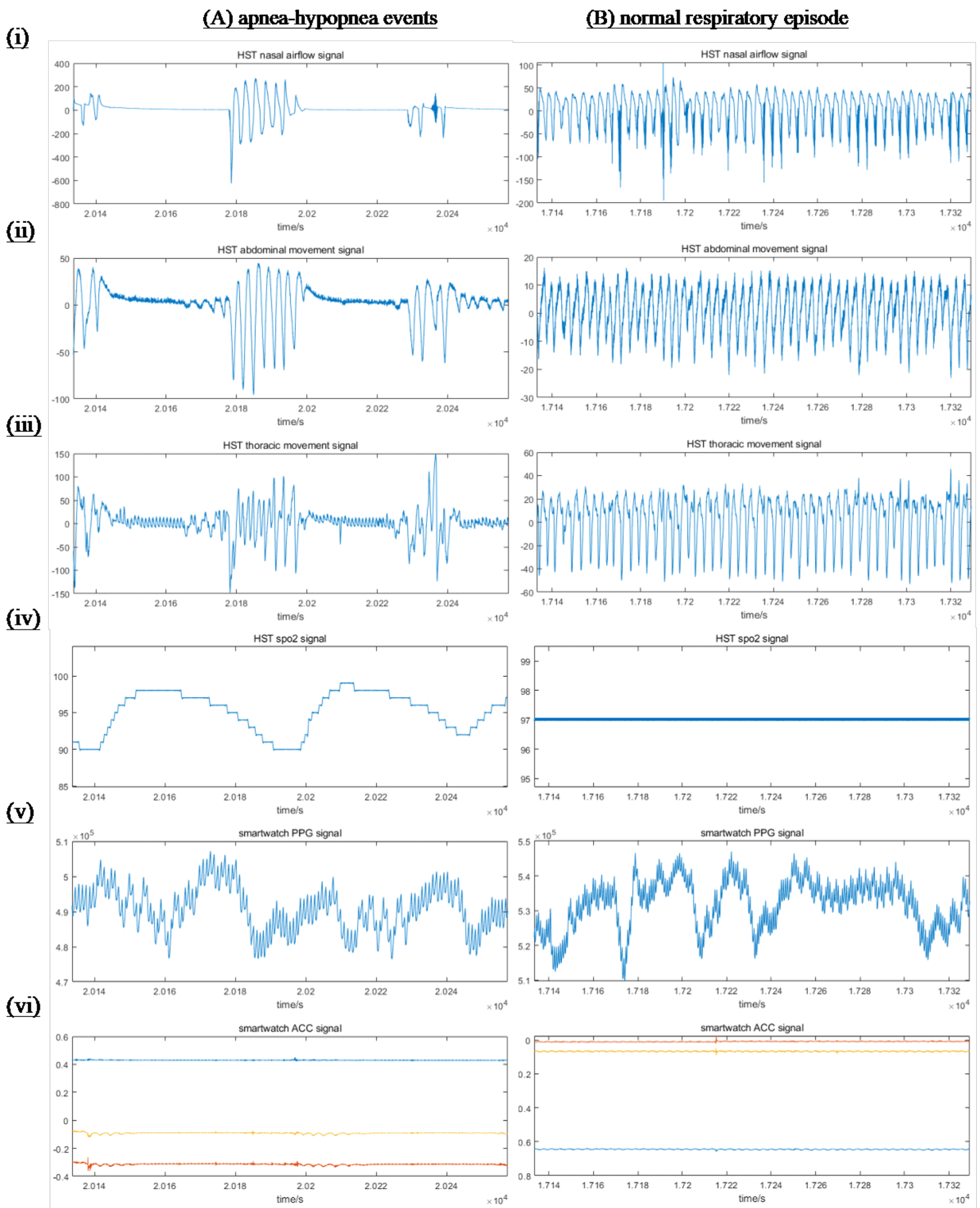
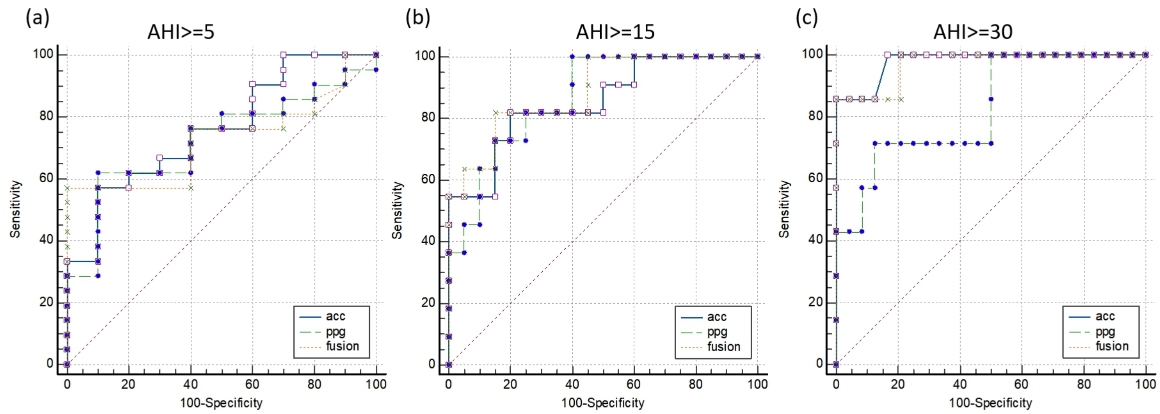


Figure 1: Monitoring signals of (i) nasal airflow, (ii) abdominal movements, (iii) thoracic movements, and (iv) blood oxygen saturation using a Home Sleep Testing (HST) device, in conjunction with signals from (v) accelerometers and (vi) photoplethysmography obtained from S7 smartwatches, in the context of (A) apnea-hypopnea events and (B) normal respiratory episodes.

a sensitivity of 0.86 and a specificity of 1.00. Based on these findings, we conclude that smartwatch-based method demonstrates substantial potential for large-scale screening of the general population. This approach may facilitate increased awareness and the adoption of beneficial lifestyle modifications among individuals with SA.

Acknowledgments

We express our deepest appreciation to Dr. Jin Guanglin, Dr. Chen Jiangxian, Dr. Zhu Bixia, and Dr. Chen Ruirui for their indispensable clinical support. This research was funded by SKG Health Technologies Co.



Methods	AUC	SE	95%CI
acc	0.762	0.0903	0.576 to 0.896
ppg	0.724	0.0944	0.535 to 0.868
fusion	0.726	0.0912	0.537 to 0.870

Methods	AUC	SE	95%CI
acc	0.855	0.074	0.682 to 0.955
ppg	0.868	0.0639	0.698 to 0.962
fusion	0.886	0.0625	0.721 to 0.972

Methods	AUC	SE	95%CI
acc	0.979	0.0231	0.851 to 1.000
ppg	0.827	0.095	0.649 to 0.939
fusion	0.97	0.0321	0.837 to 0.999

Figure 2: The ROC curves corresponding to AHI thresholds of (a) 5, (b) 15, and (c) 30 were evaluated. For each threshold, models utilizing accelerometer (acc), photoplethysmogram (ppg), and a fusion of both features (acc+ppg) were analyzed separately.

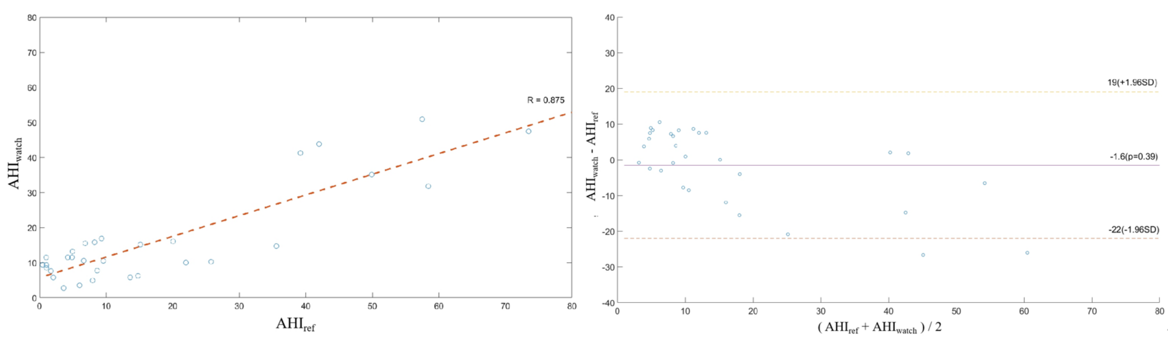


Figure 3: A Bland-Altman plot and a correlation plot were employed to compare the AHI value between the watch (AHI_{watch}) and reference (AHI_{watch}).

Conflicts of Interest

All authors are affiliated with SKG Health Technologies Co., Ltd., which manufactures the S7 smartwatches used in this study and funded the research. They disclose these affiliations to ensure transparency and acknowledge potential biases, while asserting their commitment to impartial and rigorous scientific standards.

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